

A RESPONSE THEORY FOR THE RIEMANN ZERO GAS: AMPLITUDE CONSERVATION, A THERMAL BRAGG MIRROR, AND THE TWO-SUM INTERFERENCE

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ABSTRACT. Two companion notes measured how the local statistics of Riemann zeros respond to the prime oscillations of the explicit formula: an amplitude channel $w(\tau)$ and a zero-displacement channel $v(\tau)$, functions of $\tau = \log p/L$, $L = \log \frac{t}{2\pi}$. Here we organize those measurements into a coherent response theory. (i) The two channels obey an *amplitude conservation law*, $\sqrt{w} + (\beta/2)v = 1$ with $\beta/2 \approx 1.07$, whose coefficient agrees with a phase-bookkeeping ratio within its extrapolation uncertainty; the law survives an out-of-sample test on unseen primes, a split-half common-mode test, and a free-exponent test consistent with $k = 2$. (ii) Extending the measurements to $\tau = 0.61$ shows that w *changes sign* at $\tau_0 = 0.40$ — the role-crossover of the companion note — with response phases that jump from 0 to π without rotating: both channels are real everywhere, consistent with a lossless (reactive) medium. (iii) A thermal Bragg model explains the crossing: the ideal mirror sits at the Riemann–Siegel fold $\tau = 1/2$; dressing it with the *measured* zero-displacement jitter and the measured indirect-channel plateau brackets the observed zero: the thermal family places it in $[0.37, 0.41]$ around the measured 0.40, with unit coherent amplitude ($A \approx 1$) throughout; window-to-window jitter differences absorb the previously unexplained fixed- τ scatter. The dephasing scale is the *standing-wave* factor: the empirical characteristic-function ratio gives $s = 4.11 \pm 0.04$ against the Gaussian value 4. (iv) The structure factor $(1 - 2\tau)$ is derived in three lines as the pair count of the folded main sum, vanishing sharply at the fold; pointwise $|Z|^2$ regressions instead follow the shifted-second-moment family $\approx 2(1 - \tau)$ through and beyond the fold, so that the complementary *mirror channel* — measurement minus the direct pairs — is a *tent* peaked at the fold — the approximate functional equation made audible as an interference channel. (v) The residual variance decomposes as $\sim 5\%$ unstripped tail, 25–45% gap-modulated prime channel, and a core growing at 0.035 per unit L — the CUE rate at one tenth the amplitude.

1. INTRODUCTION

The first companion note [5] measured the joint law of consecutive-zero gaps and between-zero maxima of Hardy’s Z -function and its deviation from CUE characteristic polynomials. The second [6] dissected that deviation into explicit-formula channels: each prime wave $\cos(t \log p)$ enters the amplitudes at full weight and splits between a direct amplitude channel w and a zero-displacement channel v , both collapsing onto one-variable laws in the Berry variable $\tau = \log p/L$ [2]; the channel decomposition itself is an observable-level counterpart of the hybrid Euler–Hadamard product [3]. This note asks what *physics* organizes those curves, and answers with five measured structures: a conservation law, a sign-crossing, a thermal Bragg model that brackets the crossing’s location, an arithmetic derivation of the structure factor, and the measurement of the complementary mirror channel. Throughout, the style of the companions is kept: no closed forms are fitted; every ingredient is either measured or derived; and every claim is accompanied by the test that tried to kill it.

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Data and conventions are those of [5, 6]: eleven amplitude windows spanning $t = 1.2 \times 10^5$ to 2.7×10^{11} (the last from Odlyzko’s table of zeros near the 10^{12} -th [4]), plus six low windows down to $t \approx 1.2 \times 10^3$ for the extended- τ measurements; channels are measured by linear regression of $\log(\text{unfolded maxima})$ and $\log(\text{unfolded gaps})$ on $\cos(t_n \log p), \sin(t_n \log p)$ at gap midpoints, with own-gap controls for w .

2. THE AMPLITUDE CONSERVATION LAW

The second companion’s sum rule — total prime content of unit weight — holds at first order in v . Its violation $g = w + \beta v - 1$ is zero at small τ and grows as $g \approx 1.17 v^2$; and $1.17 \approx \beta^2/4$ with $\beta = 2.14$. The square completes:

$$w = \left(1 - \frac{\beta}{2}v\right)^2 \quad \Longleftrightarrow \quad \sqrt{w} + \frac{\beta}{2}v = 1.$$

The coefficient has an independent motivation: since a gap is the derivative of the counting function, the displacement response $U = v/\tau$ is the quadrature partner of the in-phase response w , suggesting $\beta = U(0)/w(0) = 2.01/0.94 \approx 2.14$. The $w(0)$ extrapolation carries the $\pm 5\%$ model uncertainty documented in [6], and the unconstrained fit below prefers a slope 1.029 ± 0.012 (window-block bootstrap), in mild ($\approx 3\sigma$) tension with $\beta/2 = 1.07$; we treat the phase-bookkeeping value as motivated, not derived. (The coarser log–log regression estimate quoted as $\beta \approx 2.3$ in [6] is the same quantity under a different convention.)

Observation 1 (Amplitude conservation). *Across 44 matched (w, v) pairs (4 primes $\times$ 11 windows, t from 10^5 to 2.7×10^{11}), the free fit gives $\sqrt{w} = 0.990 - 1.025 v$ with residual RMS 0.0096 (slope 1.029 ± 0.012 under window-block bootstrap), at the irreducible per-prime scatter floor; constraining the intercept to 1 gives slope $1.069 \approx \beta/2$.*

Three attempts to kill the law failed (Figure 1). *Out of sample*: w for $p = 11, 13$ — never used in deriving the law, and extending τ to 0.26 — follows the line with RMS 0.0148 over 22 new pairs (with a small one-sided mean offset of -0.009) while $\sqrt{w}$ falls from 0.78 to 0.45. *Common mode*: measuring w from even-indexed gaps and v from odd-indexed gaps (independent noise) reproduces the law with $a = 0.998$, $b = 1.051$, RMS 0.019 against 0.018 for same-half pairs. *Free exponent*: fitting $w = (1 - bv)^k$ jointly is degenerate along a (b, k) valley (window-block bootstrap spans $k \approx 2$ –4 with compensating slope), so the exponent alone is determined only at $O(1)$; conditioning b on the linear law gives $k = 2.01$, and the discriminating test is the form race: among equal-parameter alternatives the intensity-unitarity form $w + cv^2 = 1$ fails by a factor ≈ 9 in RMS. What is conserved is the *amplitude*, not the intensity: the prime wave’s unit amplitude splits linearly between the transmitted amplitude $\sqrt{w}$ and the absorbed displacement $(\beta/2)v$.

3. THE SIGN CROSSING AND LOSSLESSNESS

Extending the channel measurements to the primes up to 31 (eleven primes) over twelve windows (132 points, τ up to 0.61) shows that the amplitude channel does not merely shrink — it *crosses zero* at $\tau_0 = 0.40 \pm 0.02$ and settles on a negative plateau $B = -0.16$: beyond the crossing, peaks are slightly *low* where the prime wave is high. The crossing coincides with the role-crossover $\tau^* \approx 0.40$ found by a different method (the exhaustion analysis of [6]), which is thereby identified: τ^* is the zero of w . Two qualifications: the plateau value depends on the regression basis ($B = -0.16$ for the eleven-prime basis used throughout; restricted bases

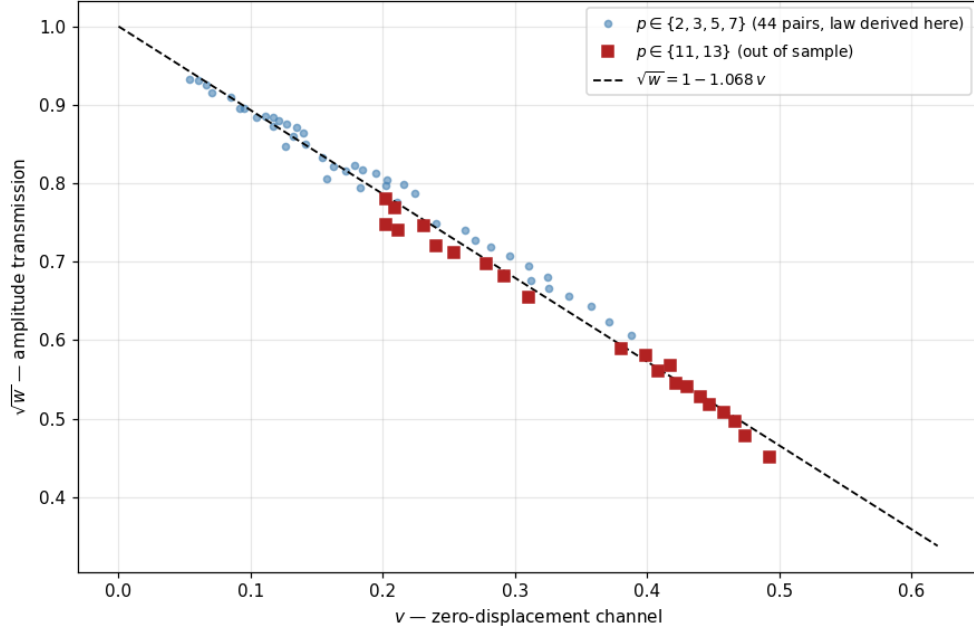


FIGURE 1. The conservation law. Blue: the 44 pairs from which the law was extracted. Red: the 22 out-of-sample pairs ($p = 11, 13$).

give larger values), and the statement “peaks are low where the wave is high” holds *at fixed observed gap* — without the own-gap controls the unconditional response is positive.

The response phases are diagnostic. Through the crossing the phase of w jumps from $\approx 0^\circ$ to $\approx 180^\circ$ with quadrature components below 3% everywhere; the phase of v stays pinned at 0° . A damped resonance would rotate through $\pm 90^\circ$; a real response that changes sign is the signature of a *lossless* (conservative, Hermitian) medium. The curtain does not absorb the prime waves; it redistributes and reflects them — consistent, at the level of response theory, with a Hermitian generator underneath. This also resolves the conservation law’s apparent paradox: w reaches zero not where v would extrapolate the law to closure, but at the sign crossing; the law describes the below-crossing branch.

4. THE THERMAL BRAGG MIRROR

Where should the crossing be? The Riemann–Siegel main sum contains only $n \leq N = \sqrt{t/2\pi}$, i.e. $\log n \leq L/2$: the amplitude has a sharp spectral fold at $\tau = 1/2$, and a wave whose period matches two lattice spacings ($1/\tau = 2$) is precisely the Bragg condition of the zero lattice. The ideal mirror sits at $\tau = 1/2$; the measured crossing sits at 0.40. The shift is thermal.

Model: $w(\tau) = A(1 - 2\tau)e^{-c\tau^2}[\tau < 1/2] + B$, where $(1 - 2\tau)$ is the ideal structure factor (derived in Section 5), B is the indirect (zero-mediated) channel measured as the beyond-fold plateau, and the Gaussian factor is Debye–Waller dephasing by the zeros’ displacement jitter. The jitter is *measured*: comparing gap midpoints with the Riemann–von Mangoldt flow gives σ_u per window; in unfolded units it grows from 0.198 to 0.272 across our windows — the crystal warms logarithmically, as the classical $S(t)$ variance suggests — giving $c = (L\sigma_u)^2/2 = 0.77\text{--}1.46$, mean 1.17.

Observation 2 (The crossing is bracketed). *With c fixed from the measured jitter and B fixed from the measured plateau, the single remaining parameter fits to $A = 1.03 \approx 1$ (unit coherent amplitude) and the model's zero lands at $\tau = 0.406$ (Figure 2); with the empirical standing-wave damping, which fits the full curve best (RMS 0.029 vs 0.052), the zero moves to 0.368; intermediate damping scales fit equally well and cross at 0.39. The thermal family thus brackets the measured crossing within $[0.37, 0.41]$; no member is simultaneously the best global fit and the exact crossing, which we attribute to the crude ideal structure factor and to partial reflection below the Bragg condition.*

Two refinements. First, giving each window its own measured $c(L)$, together with one fitted overall damping scale, collapses the previously unexplained fixed- τ , window-to-window scatter of the channel measurements from ± 0.04 to ± 0.003 : the “secondary variable” of the companion note is the window temperature. Second, the *scale* of the dephasing is consistent with standing-wave physics: at a Bragg mirror the intracavity field has period $2k$, so dephasing involves e^{2iku} rather than e^{iku} . Measured empirically — as the ratio of log characteristic functions of the displacement series, with no Gaussian assumption — this gives

$$s_{\text{eff}} = \frac{\ln |\langle e^{2iku} \rangle|}{\ln |\langle e^{iku} \rangle|} = 4.11 \pm 0.04,$$

the Gaussian value 4 dressed by the measured sub-Gaussian displacement statistics (excess kurtosis -0.3 to -0.5 — level repulsion suppressing the tails). Replacing the Gaussian factor by the empirical $|\langle e^{2iku} \rangle|$ halves the model residual (RMS $0.074 \rightarrow 0.052 \rightarrow 0.029$ for no damping, naive, and standing-wave damping respectively). Two honest caveats: the ratio s_{eff} is largely a measurement of the near-Gaussianity of u (any near-Gaussian series gives ≈ 4), and damping at intermediate scales (e.g. $1.5k$) fits the w curve equally well while crossing at 0.39; so the standing-wave scale is *consistent* rather than demonstrated, with partial reflection below the Bragg condition the natural interpolation — an open refinement.

5. THE STRUCTURE FACTOR, DERIVED

Derivation 3. Write $Z = 2 \sum_{n \leq N} n^{-1/2} \cos \phi_n$, $\phi_n = \theta(t) - t \log n$. In $|Z|^2$ the frequency $\log p$ is produced exactly by the pairs (m, mp) with $mp \leq N$, each contributing weight $4(m \cdot mp)^{-1/2}$, while the DC part is $2 \sum_{n \leq N} n^{-1}$. Hence the relative modulation is

$$R_{\text{direct}}(p) = 2p^{-1/2} \frac{S_1(N/p)}{S_1(N)}, \quad S_1(x) = \sum_{m \leq x} \frac{1}{m},$$

which is $p^{-1/2} \cdot 2(1 - 2\tau)$ in the continuum limit and vanishes identically for $p > N$.

The structure factor $(1 - 2\tau)$ of the Bragg model is therefore not a guess: it is the pair-counting arithmetic of the folded sum, with a sharp horizon at the fold.

6. THE MIRROR CHANNEL: A TENT AT THE FOLD

Testing the derivation pointwise — regressing $|Z(t)|^2$ on the prime waves over fine grids (1.3×10^6 points per window, two windows) — produced a surprise: the measured modulation does *not* vanish at the fold. It follows, through and beyond the fold, the shifted-second-moment family

$$R(p) \approx 2p^{-1/2} \frac{L - \log p + c_0}{L + c_0} \approx p^{-1/2} \cdot 2(1 - \tau),$$

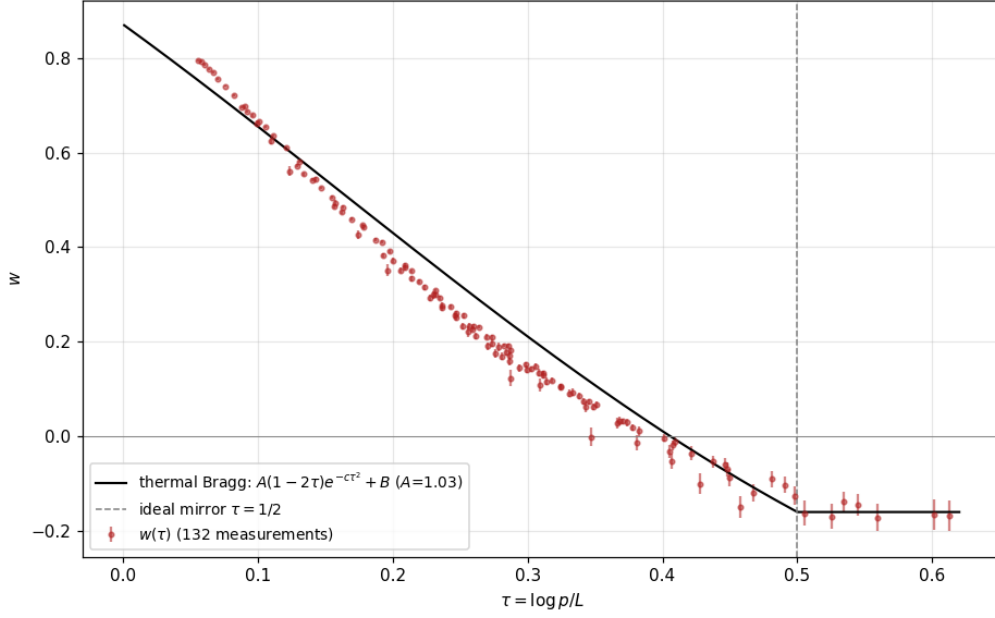


FIGURE 2. The sign crossing and the thermal Bragg model with c from the measured jitter, B from the measured plateau, and $A = 1.03$ the only fitted number.

dying only at the full horizon $p \approx t/2\pi$ (cf. the classical mean-value theorems for $\int |\zeta|^2(m/n)^{it}$ [1]). The difference

$$R_{\text{mirror}} = R_{\text{measured}} - R_{\text{direct}}$$

is a *tent*: it rises as 2τ below the fold, peaks at the fold with value 1, and descends as $2(1 - \tau)$ beyond (Figure 3). Its mechanism is the chirp cross-terms $\cos(2\theta(t) - t \log nm)$ — precisely the interference between the two mirror-image sums of the approximate functional equation — which lock onto frequency $\log p$ at the stationary-phase products $nm \approx (t/2\pi)/p$. Beyond the fold, where no direct pairs exist, everything one hears is the mirrored sum: the functional equation of ζ , measured as an interference channel. The parameter-free tent sits systematically below the two-window measurements by 6–20%; the exact normalization awaits the true main-term constants (the free fit prefers $c_0 \approx 1.1$ over the naive $2\gamma - 1$). Measurements at $\tau \gtrsim 0.62$ are excluded: the zero-spacing comb band sets in there, and non-prime control frequencies reach the tent scale — unlike the max-channel measurements, these pointwise regressions do not yet carry a full placebo battery, which we flag as a limitation.

7. THE RESIDUAL VARIANCE, DECOMPOSED

The companion note’s residual amplitude variance $V_{\text{res}}(L) = 0.163 \rightarrow 0.244$ (after stripping all $p^k \leq 300$) decomposes as follows. The unstripped tail, computed from the measured $w(\tau)$ curve including its negative plateau, contributes only $\sim 4.5\%$. Re-stripping with gap-interaction terms (the transmission is gap-dependent) removes, placebo-corrected, a further $0.074 \rightarrow 0.062$ — 45% to 25% of the residual — and lifts the stripped core correlation to 0.984. What remains grows linearly at 0.035 per unit L : within errors the *CUE amplitude-variance rate* (0.036 per unit N), at one tenth the magnitude (the companion’s “one sixth” referred

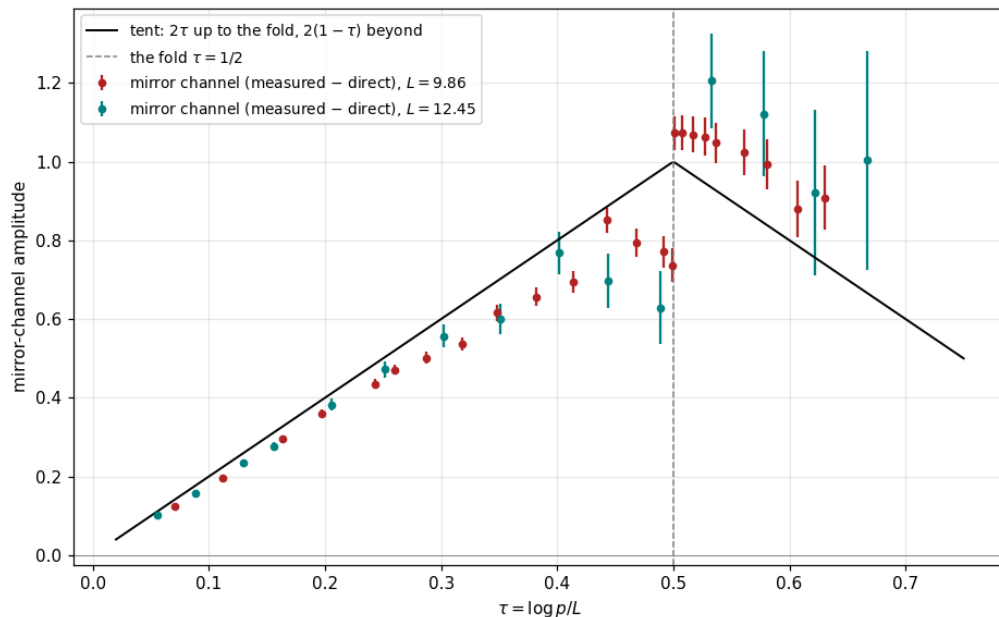


FIGURE 3. The mirror channel at two windows against the parameter-free tent. The direct channel dies at the fold; the mirror channel peaks there.

to the raw, undecomposed residual). The zero gas carries an intrinsic noise component that beats at the random-matrix tempo but at a whisper.

8. SYNTHESIS AND OPEN PROBLEMS

The measured response theory of the zero gas now reads: each prime wave enters at unit amplitude and is split, conserving amplitude, between transmission and zero displacement ($\sqrt{w} + (\beta/2)v = 1$); the medium is lossless, with a Bragg mirror at the Riemann–Siegel fold whose observed position is shifted to 0.40 by the measured, logarithmically growing thermal jitter, with standing-wave dephasing of measured scale 4.11; the ideal structure factor is the fold’s pair-counting arithmetic; beyond the fold the response is carried entirely by the functional equation’s mirror sum, whose interference channel is a tent peaked at the fold; and the last residue of randomness beats at the CUE rate at one tenth amplitude.

Open problems: (1) the exact tent normalization from the classical shifted-moment main terms, including the χ -side; (2) the mapping of the pointwise mirror channel to the max-channel plateau $B = -0.16$ (sign and size); (3) partial-reflection dressing $r(\tau)$ below the Bragg condition; (4) a derivation of β ; (5) the distribution (not just variance) of the intrinsic core — is it GUE?; (6) universality: does the same response theory, with the same constants, hold for Dirichlet L -functions?

REPRODUCIBILITY

All measurements are scripts 62–78 in <https://github.com/azizran/deney> (directory `qm_riemann/`), runnable from the stored datasets; the dated research log records the failed intermediate models alongside the results.

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